

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mining Engineering

2. Subject Name: Surface Mining

L-T-P: 3-0-0

Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Week 1: Introduction to Surface Mining, Current status and challenges, Understanding rocks, minerals and deposits

Week 2: Cut off grade, Surface mining phases and surface mining systems, Surface mining, Excavations and unit operations.

Week 3: Drilling and blasting for surface mines

Week 4: Shovel and dumper operation for surface mining excavations

Week 5: Operation of Dozer and dragline

Week 6: Operation of continuous surface miner

Week 7: Operation of bucket wheel excavator and bucket chain excavator

Week 8: Transportation system for surface mines and expit transportation

Week 9: Design and maintenance of haul roads

Week 10: Stability assessment of pit slopes and dump designs

Week 11: Sea bed mining, dimensional stone mining

Week 12: Safety Environment and mine closure

Reference Books:

1. Mishra G.B.,1978, Surface Mining, Dhanbad Publishers

2. Das S.K.,1998, Surface Mining Technology, Lovely Prakashan

3. Deshmukh R.T., 1996, Opencast Mining, M. Publications, Nagpur.

4. De Amithosh,1995, Latest Development of Heavy Earth Moving Machinery, Annapurna Publishers

5. Hartman H.L., 2002, Introductory Mining Engineering, Publishers John Willey and sons.

6.Peter Darling, 2011, SME Handbook, SME Publication

7. Rzhovsky, V.V.,(1983),Opencast Mining Unit. Operation, Mir publications.

8.Rzhovsky, V.V.,(1985), Opencast Mining Technology and Integrated Mechanisations, Mir publications.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Surface Mining, Current status and challenges, Understanding rocks, minerals and deposits	3
2.	Cut off grade, Surface mining phases and surface mining systems, Surface mining, Excavations and unit operations.	3
3.	Drilling and blasting for surface mines	3
4.	Shovel and dumper operation for surface mining excavations	3
5.	Operation of Dozer	3
6.	Operation of Dragline	3
7.	Operation of continuous surface miner	3
8.	Operation of bucket wheel excavator and bucket chain excavator	3
9.	Transportation system for surface mines and expit transportation	3
10.	Design and maintenance of haul roads	3
11.	Sea bed mining, dimensional stone mining	3
12.	Safety Environment and mine closure	3
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Total number of hours		36